

MATÍAS FERNÁNDEZ LAKATOS

Data Analyst · Data Scientist · PhD

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· [ORCID](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

Data analyst and scientist with 10+ years turning complex, high-dimensional data into decisions. MSc in Big Data Analytics (data mining, visualization, business intelligence) on top of a PhD in Physics — rigorous statistics with a practical, stakeholder-oriented mindset. Currently **Senior Neuromorphic Researcher at Gradient**; previously analyzed large-scale behavioral data streams for cybersecurity, defining metrics, calibrating decision thresholds, and communicating model results to non-technical audiences through **explainable AI** and clear technical documentation. Strong **Python (pandas/NumPy)**, **SQL**, **statistics**, and **visualization**. Neo4j Certified Professional.

PROFESSIONAL EXPERIENCE

Senior Neuromorphic Researcher

Gradient · Vigo, Spain · *Jul 2026 – present*

- Conducting research on neuromorphic computing (Spiking Neural Networks) applied to the analysis of different data types, with emphasis on **images and time series**.
- Producing and maintaining **state-of-the-art reviews and technical documentation**: analyzing new publications, hardware devices, and software tools, and contextualizing their relevance against project objectives.
- Evaluating potential **real-world applications and use cases** (autonomous vehicles, manufacturing, healthcare) for neuromorphic systems, including energy-efficiency trade-offs for edge AI.

Research Engineer (Data Science)

Gradient · Vigo, Spain · *2025 – Jun 2026*

- Analyzed **large-scale behavioral data streams (~300K events)** for early threat detection in corporate cybersecurity.
- Defined **evaluation frameworks and metrics**; calibrated decision thresholds via percentile-based selection with G-mean and loss-based criteria.
- Made model results interpretable for **non-technical users** with explainability techniques (SHAP/XAI); authored technical documentation for stakeholders (INCIBE).
- Tracked experiments and versioned models with **MLflow**; worked within a **Kafka/Spark/Airflow** data architecture.

Researcher, Lecturer & Teaching Assistant

Faculty of Engineering, Universidad de la República (Udelar) · Montevideo, Uruguay · *2015 – 2024*

- Designed **data-processing pipelines and statistical analyses** for experimental physics data in high-dimensional settings.
- Published **6 peer-reviewed articles** (4 first-author, Scopus); presented results at international conferences — strong data storytelling and technical communication.
- Taught university-level courses with heavy statistical content; supervised students in quantitative modeling.

TECHNICAL SKILLS

Data Analysis & Visualization

[pandas](#) · [NumPy](#) · [SciPy](#) · [matplotlib](#) · [seaborn](#) · [plotly](#) Exploratory Data Analysis · Statistical Analysis · Multivariate Analysis · Hypothesis Testing · Time Series · Signal Processing · Data Storytelling

Databases & Querying

[SQL](#) (relational) · [Neo4j](#) / [Cypher](#) (Certified Professional) · [MinIO](#) (object storage) Data Modeling · Query Optimization · Large-Scale Databases

Programming & Tools

[Python](#) · [R](#) · [MATLAB](#) · [LaTeX](#) · [Git](#) · [Linux](#) · [Excel](#)

Machine Learning (working knowledge for analysis)

[scikit-learn](#) · [TensorFlow/Keras](#) · [MLflow](#) · Anomaly Detection · Explainable AI (SHAP) · Model Evaluation & Validation

Business & Big Data (MSc coursework)

Data Mining · Data Visualization · Business Intelligence · Business Applications & Use Cases · IoT Data · Kafka / Spark / Airflow

EDUCATION

- **MSc in Big Data Analytics Technologies** — Universidad de Santiago de Compostela, Spain (2024–2025). Thesis: *Application of new anomaly detection models in Big Data contexts*. Coursework: Large-Scale Databases, Data Mining, Data Visualization, Business Intelligence, Statistics, IoT, Business Applications and Use Cases.
- **Ph.D. in Optics (Physics)** — Universidad de la República, Uruguay (2019–2024). Coursework: Computational Multivariate Statistics, Computer Image Processing, Reinforcement Learning.
- **MSc in Theoretical Physics (QCD)** — Universidad de la República, Uruguay (2016–2018).
- **BSc in Physics** — Universidad de la República, Uruguay (2011–2015).

CERTIFICATIONS

- **Neo4j Certified Professional** — Neo4j GraphAcademy, May 2026, plus 6 GraphAcademy courses (Cypher, data modeling, data import — [full profile](#))
- **Master Python for DevOps and CI/CD** — Udemy, 2025

SELECTED PUBLICATIONS

6 peer-reviewed publications (4 first-author, Scopus-indexed) in optics, imaging, and theoretical physics — full list on [ORCID](#).

LANGUAGES

Spanish — Native · **English** — Professional working proficiency

Last updated: July 2026